Task- Session 1

Name:Hamid Reza Ghahramani

META-ANALYSIS

YOUR ROLE

You have just joined a research team as a junior research assistant. A preliminary meta-analysis project has been prepared for you, but several decisions made during the process need to be checked. Your job is to review the evidence, choose appropriate approaches and interpret reported results.

IMPORTANT

This task is designed to assess your understanding of the concepts introduced in the session. You are NOT required to perform any numerical calculations. All numerical results needed for interpretation are already provided. Your job is to explain what those numbers mean and justify your decisions.

Before you start:

- Read each case carefully before answering.
- Do not focus only on selecting the correct option. Your reasoning matters.
- When asked to interpret a number, do not recalculate it. Explain what the reported value tells you.

Your mission:

A research team wants to investigate whether a cognitive intervention improves working-memory performance in adults. Several studies have been collected, but they do not all look the same. Some use different populations, interventions, or outcomes, and the results are reported in different forms.

Your task is to examine the project step by step. You will first decide which studies are relevant, then choose appropriate effect-size measures, interpret reported effect sizes, and finally evaluate a diagnostic-accuracy case.

The research question is:

RESEARCH QUESTION

Among adults, does a cognitive training intervention improve working-memory performance compared with a control condition?

PART 1 — Study Selection: What Belongs in the Meta-analysis?

The previous researcher identified the following studies:

Study	Population	Intervention	Outcome	Sample size
A	Adults	Cognitive training	Working memory	80
B	Adults	Cognitive training	Working memory	65
C	Children	Cognitive training	Working memory	120
D	Adults	Medication	Working memory	90
E	Adults	Cognitive training	Anxiety	75
F	Adults	Cognitive training	Working memory	40
G	Adults	Cognitive training	Working memory	500
H	Adults	Cognitive training	Working memory	55

1. **Which studies would you keep for the meta-analysis and <u>why</u>?**

I would keep Studies A, B, F, G, and H for the meta-analysis, assuming that they include a control condition. First, we define the PICO elements based on the research question: the Population is adults, the Intervention is cognitive training, the Comparator is a control condition, and the Outcome is working-memory performance. We then screen the studies and compare their characteristics with the predefined PICO criteria to determine which studies are eligible for the meta-analysis.

2. **Which studies would you exclude and <u>why</u>?**

I would exclude Study C because the population is children, Study D because the intervention is medication, and Study E because the outcome is anxiety rather than working-memory performance.

SMD can be used if the included studies measure working-memory performance using different scales or tests. However, SMD does not make studies with different populations, interventions, or outcomes eligible for the meta-analysis.

3. **Study G has the largest sample size. Does its large sample size, by itself, make it more appropriate for inclusion? <u>Explain your reasons.</u>**

No. A large sample size by itself does not make Study G more appropriate for inclusion in the meta-analysis. A study must first meet the predefined eligibility criteria, including the PICO elements and other study-selection criteria.

However, if Study G meets these criteria, its large sample size can increase the precision of its effect estimate. A larger sample size generally leads to a smaller standard error and therefore gives the study greater statistical weight in the meta-analysis. However, sample size alone should not determine whether a study is eligible for inclusion.

PART 2 — Choosing the Appropriate Effect Size + Interpret your answers

Possible measures:

• Mean Difference (MD) • Standardized Mean Difference (SMD) • Odds Ratio (OR) • Risk Ratio (RR) • Risk Difference (RD)

1. **CASE A:** Two groups are compared on working-memory performance, and both groups are assessed with the same measurement scale.
2. **CASE B:** Two studies assess the same general outcome, but they use different measurement scales for that outcome.
3. **CASE C:** Participants are classified as either having improved or not having improved after the intervention.
4. **CASE D:** A study reports whether an event occurred in the intervention group and in the control group, and the researcher wants to compare the occurrence of that event between groups.

Case A — Mean Difference (MD)

I would use the **Mean Difference (MD)** because both groups are assessed using the **same measurement scale** for working-memory performance. MD directly compares the difference between the two group means.

Case B — Standardized Mean Difference (SMD)

I would use the **Standardized Mean Difference (SMD)** because the studies assess the same general outcome but use **different measurement scales**. SMD standardizes the results so that the effect sizes can be compared and combined in a meta-analysis.

Case C — Odds Ratio (OR), Risk Ratio (RR), or Risk Difference (RD)

This is a **dichotomous outcome** because participants are classified as either improved or not improved. Therefore, OR, RR, or RD can be used depending on what measure the researcher wants to compare. For example, if the researcher wants to compare the **odds of improvement**, OR would be appropriate.

Case D — Odds Ratio (OR), Risk Ratio (RR), or Risk Difference (RD)

This is also a **dichotomous outcome**, because the study records whether an event occurred or did not occur. Therefore, OR, RR, or RD can be used. If the researcher wants to compare the **odds of the event occurring** between the intervention and control groups, OR would be appropriate.

PART 3 — Interpret Reported Effect Sizes

1. **Mean Difference (MD) = +4**
2. **Standardized Mean Difference (SMD) = +0.45**
3. **Odds Ratio (OR) = 2.10**
4. **Risk Ratio (RR) = 1.50**

1. Mean Difference (MD) = +4

The intervention group had a mean outcome that was **4 units higher** than the control group. Because MD uses the original measurement scale, the effect can be interpreted directly in the units of the outcome.

2. Standardized Mean Difference (SMD) = +0.45

The intervention group had a mean outcome that was **0.45 standard deviations higher** than the control group. SMD is commonly used when the same continuous outcome is measured using different scales or tests across studies.

3. Odds Ratio (OR) = 2.10

The **odds of the event occurring** in the intervention group were **2.10 times the odds** in the control group. If the event is improvement, this means the odds of improvement were 2.10 times higher with the intervention.

4. Risk Ratio (RR) = 1.50

The **risk or probability of the event occurring** in the intervention group was **1.50 times the risk** in the control group. This represents a **50% higher relative risk** of the event in the intervention group.

PART 4 — Diagnostic Accuracy: Read the Evidence (diagnostic/prognostic studies)

KEY TERMS

True Positive (TP): the test is positive, and the reference standard indicates the condition is present.
False Positive (FP): the test is positive, but the reference standard indicates the condition is absent.
False Negative (FN): the test is negative, but the reference standard indicates the condition is present.
True Negative (TN): the test is negative, and the reference standard indicates the condition is absent.

	Reference standard: condition present	Reference standard: condition absent
Index test: positive	72	18
Index test: negative	8	102

1. **Identify TP, FP, FN, and TN from the table above**.

True Positive (TP) = 72

False Positive (FP) = 18

False Negative (FN) = 8

True Negative (TN) = 102

2. **The team has already calculated the diagnostic indices: Sensitivity = 90%; Specificity = 85%.**
   What does the sensitivity value tell you about the test?
   What does the specificity value tell you about the test?
   Sensitivity = 90%:
   This means that the test correctly identifies **90% of people who actually have the condition**. In other words, among people with the condition, 90% receive a positive test result.
   Specificity = 85%:
   This means that the test correctly identifies **85% of people who do not have the condition**. In other words, among people without the condition, 85% receive a negative test result.